

Bengali ASR: seven models on FLEURS, one GPU

Third report · 920 FLEURS bn_in TEST utterances · RTX A5000 · greedy, batch 1, no external LM · all seven re-measured in one sitting

Model	WER	95% CI (clustered)	CER	ms/clip	Params
Conformer Large (hishab)	14.37%	[13.11-15.73]	4.48%	77	121.5M
Whisper Medium	16.23%	[15.05-17.45]	5.09%	1,866	763.9M
Qwen3-ASR + Bengali adapter	16.54%	[15.43-17.68]	4.68%	9,728	2.04B / 38M trained
► ehzawad FastConformer-CTC (Phase 4)	17.12%	[15.95-18.34]	5.12%	83	115.6M
ehzawad FastConformer (Phase 1)	19.67%	[18.50-20.88]	5.79%	76	115.6M
Wav2Vec2	20.64%	[19.46-21.89]	6.02%	64	315.5M
hishab FastConformer	36.22%	[33.88-38.72]	9.05%	79	115.6M

The new row is a THIRD decode of FLEURS test for the Phase-4 lineage: an owner-authorised interim look at update 88,000 (17.54%) and the registered final evaluation of the selected checkpoint (17.1162%, its pipeline's frozen scorer) came first. This is a post-release reproduction under this benchmark's scorer, which agrees at displayed precision - not a fresh held-out measurement. Validation-split contamination via fleurs_train applies to Phase 4 too, so the table is the 920-utterance test split. The 1,692 h corpus passed the pipeline's registered FLEURS-test overlap checks (text key and audio fingerprint; zero matches retained).

Reading this table

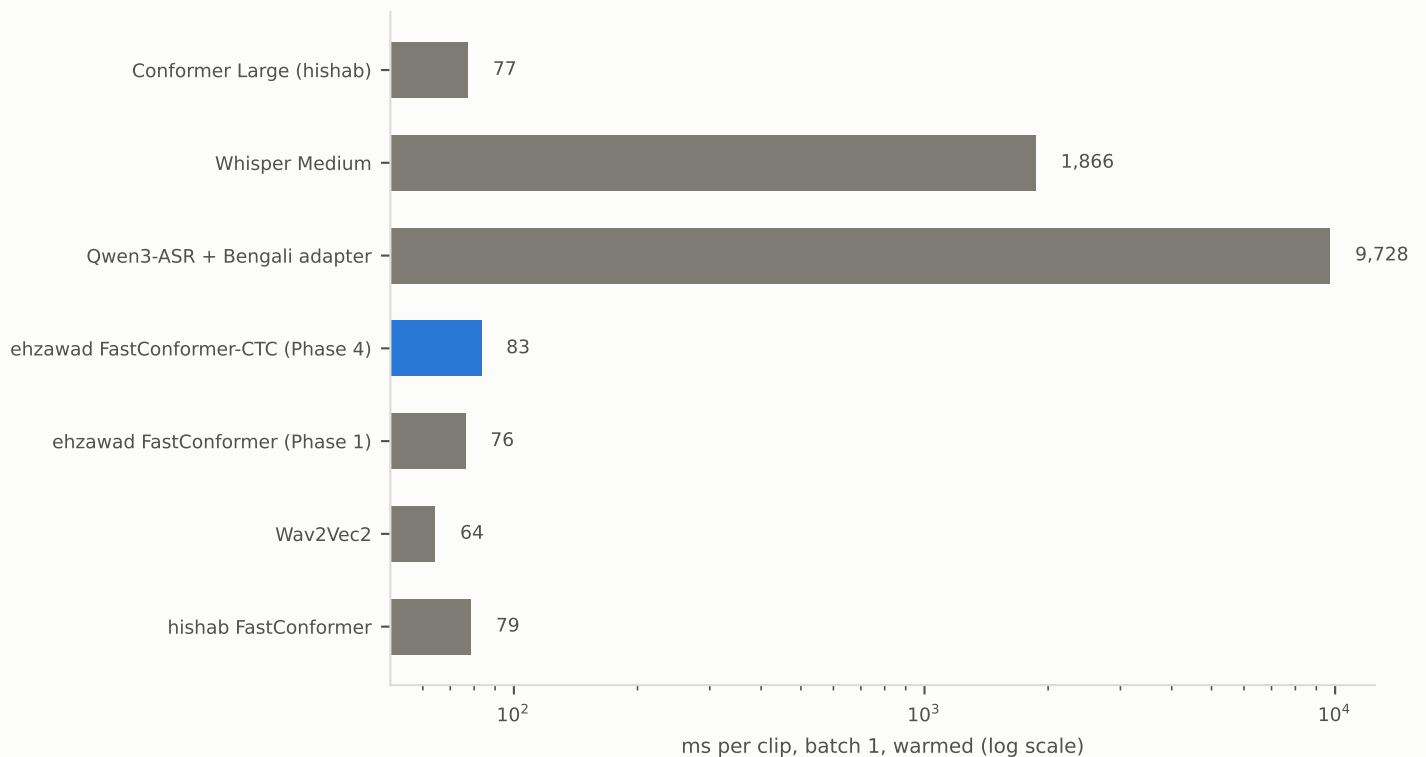
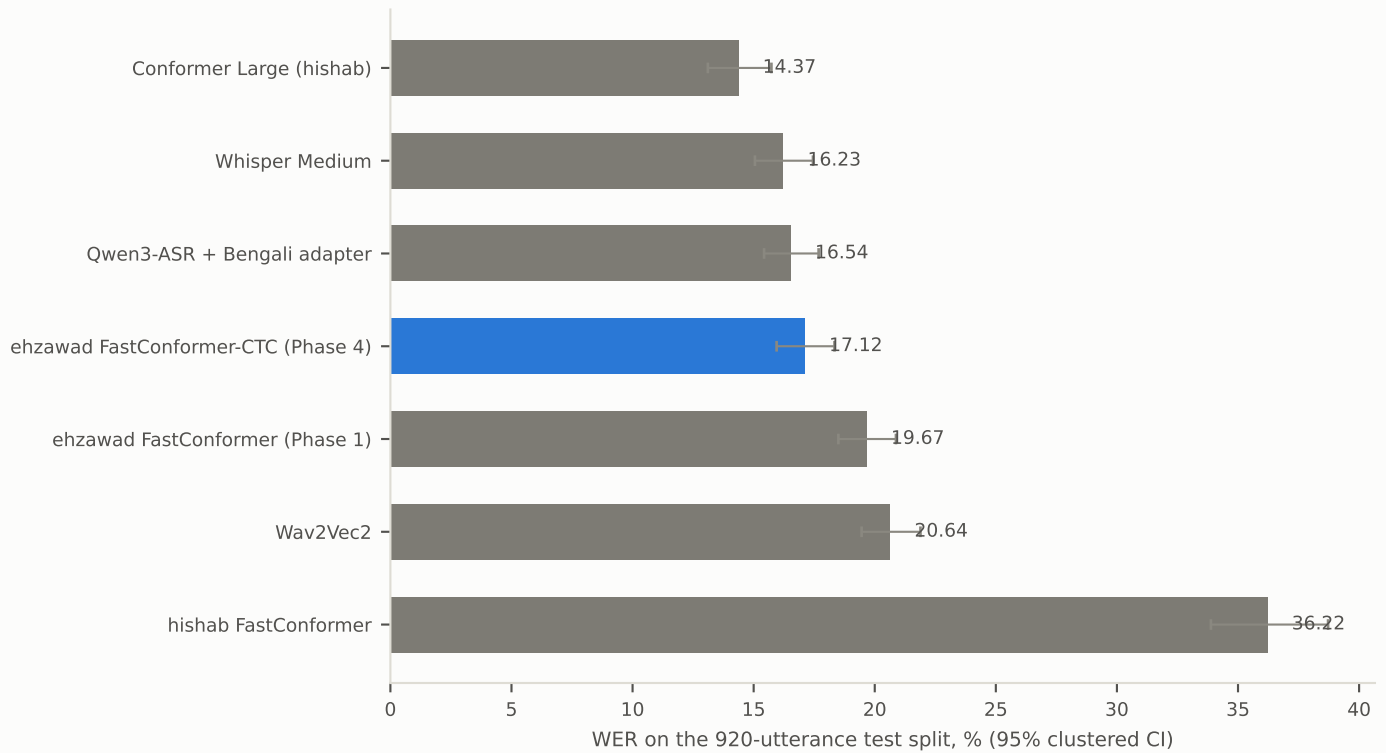
ehzawad/stt_bn_fastconformer_ctc is the Phase-4 successor to ehzawad/stt_bn_fastconformer: the same FastConformer-CTC large retrained on 1,692 hours of human-supervised Bengali (Phase 1 used 984), CC-BY-SA-4.0. It takes Phase 1's place; Phase 1 stays as a labelled predecessor row because both are public and the gap between them is itself a result. The six previously published models reproduce their second-report test-split numbers to the digit (19.67, 14.37, 36.22, 16.23, 20.64, 16.54), which is the evidence that nothing in the harness moved.

Paired sentence-clustered deltas for the new model (10,000 draws):

- vs Phase 1 -2.56 pp [-3.12, -2.01]
- vs Qwen3 adapter +0.58 pp [-0.24, +1.40]
- vs Whisper Medium +0.89 pp [+0.27, +1.50]
- vs Conformer Large +2.74 pp [+2.01, +3.46]

A clear 2.6-point improvement over its predecessor at the same size and speed; not distinguishable from the Qwen3 adapter (which it was registered to beat and did not); under a point behind Whisper Medium at 22x its speed; 2.7 points behind Conformer Large. Timing caveat: a co-tenant's 425 MiB idle process was resident on the card for the first ~110 s of the Phase-4 run and left at 10:40:52 UTC; every later model ran on an empty card. Accuracy is unaffected; the 83 ms/clip is an upper bound.

FLEURS test: accuracy and speed side by side



The two orderings are not the same ordering. Conformer Large, Phase 4, Phase 1 and hishab FastConformer sit within 10% of each other in speed; Whisper Medium costs ~22x and the Qwen3 adapter ~120x per clip at batch 1 for their accuracy.

Beyond FLEURS: official test splits of four other corpora

Fourth report · frozen seeded 1,000-utterance subsets of the OFFICIAL test splits · one model at a time · same backends and scorer

Set	official rows	sampled	audio	ref. words	stratified by
Vaani test	12,695	1,000	1.26 h	10,082	29 state/district strata
SPRING-INX R1	1,879	1,000	2.54 h	19,706	Latin / no-Latin reference
SPRING-INX R2	1,872	1,000	2.61 h	20,353	Latin / no-Latin reference
Kathbath known	2,806	1,000	1.79 h	10,801	speaker gender
Kathbath unknown	1,783	1,000	1.81 h	10,887	speaker gender

Model (WER %)	Vaani	SPRING R1	SPRING R2	Kathbath kn	Kathbath unk	flags
► ehzawad FastConformer-CTC (Phase 4)	24.66	39.24	38.41	13.38	13.43	N N N S S
Whisper Medium	26.25	39.96	39.70	9.69	9.79	U U U U U
ehzawad FastConformer (Phase 1)	26.56	41.30	40.57	17.14	16.96	N N N S S
Conformer Large (hishab)	33.10	42.46	42.24	9.33	8.79	N N N S S
hishab FastConformer	28.71	46.47	45.08	25.81	29.87	U U U U U
Wav2Vec2	31.26	47.28	47.33	19.08	18.80	U U U U U

flags, per set in the column order: N = corpus not in the model's declared training data; S = trained on a sibling split of the same corpus (Kathbath train; Conformer Large also its validation split); U = training provenance not documented. None of these means "verified clean". Phase 4's Vaani and SPRING rows are its STORED one-shot hypotheses (decoded once, 2026-09-15 08:26 UTC, batch 32 in its pipeline's gated tool), re-scored on exactly these utterances - not decoded again. Its Kathbath rows are fresh batch-1 decodes here, its first contact with Kathbath test. The Qwen3 adapter was dropped from this report by the author (~10 s/clip at batch 1).

What the table says

On the two corpora nobody trained on, the FLEURS ranking does not hold. Conformer Large - the FLEURS leader at 14.4% - is last on Vaani (33.1%) and fifth on SPRING; its training list is Bangladesh-centric and these are Indian-district (Vaani) and IIT-Madras conversational (SPRING) recordings. Phase 4 leads Vaani and both SPRING sets, Whisper Medium is second on all three at ~15x the latency, and the Phase 1 to Phase 4 gain holds on every set (-1.9, -2.1, -2.2 pp).

On Kathbath the order flips back: Conformer Large (9.3 / 8.8%) and Whisper (9.7 / 9.8%) lead and Phase 4 is third (13.4 / 13.4%). Conformer Large trained on Kathbath validation as well as train, Whisper's provenance for Kathbath is unknown (a U cell can hide training overlap), and Phase 4 saw only the train split. Known- vs unknown-speaker Kathbath differ by under a point for every model except hishab FastConformer. Sentence-clustered 95% intervals for every cell are in the README and summary.

Code-switching, latency, and how the runs were controlled

SPRING-INX: WER on utterances whose reference has no Latin-script character / has Latin-script characters (R1: 499 / 501; R2: 542 / 458), and hypotheses containing any Latin script (of 1,000)

Model	R1 no-Latin / Latin	R2 no-Latin / Latin	hyps with Latin R1 / R2
► ehzawad FastConformer-CTC (Phase 4)	35.39 / 41.85	35.14 / 41.12	0 / 0
Whisper Medium	35.92 / 42.71	36.67 / 42.22	0 / 0
ehzawad FastConformer (Phase 1)	37.94 / 43.58	38.08 / 42.63	0 / 0
Conformer Large (hishab)	40.03 / 44.11	39.53 / 44.49	0 / 0
hishab FastConformer	47.32 / 45.90	43.66 / 46.26	0 / 0
Wav2Vec2	43.99 / 49.51	44.39 / 49.76	0 / 0

The NeMo models' Bengali-only vocabularies cannot emit a Latin word; Whisper and wav2vec2 could in principle, but none of the six produced a single Latin-script hypothesis in 2,000 utterances, so every Latin reference word is an error for every model and the Latin half separates the models less than the Latin-free half does. There Phase 4 (35.4 / 35.1%) and Whisper (35.9 / 36.7%) are within each other's intervals.

ms per clip, batch 1, warmed, default runtime (N/A = stored hypotheses, not timed here)

Model	Vaani	SPRING R1	SPRING R2	Kathbath kn	Kathbath unk
► ehzawad FastConformer-CTC (Phase 4)	N/A	N/A	N/A	72	70
Whisper Medium	919	1,454	1,494	1,014	1,041
ehzawad FastConformer (Phase 1)	73	75	73	70	70
Conformer Large (hishab)	74	75	76	70	71
hishab FastConformer	72	74	73	70	71
Wav2Vec2	33	49	51	35	35

Protocol. Every set is a frozen subset (seed 20260915; proportional stratification; per-file SHA-256 verified before decoding). One model resident at a time on the A5000, selected by UUID. A run refused to start while any foreign process held the card, and a watcher polled the card every 15 s during each run, ready to stop it cooperatively; no foreign process appeared during any of the 27 runs. Nothing was ever signalled. Manifests: multiset/; raw predictions and per-run GPU state: outputs_multiset/; harness bench_run_set.py (imports bench_run.py's backends unchanged); scorer score_multiset.py. Reproduce the scoring from the committed predictions.

Evaluation data: FLEURS (Google, CC-BY-4.0); Vaani transcription part (ARTPARK-IISc, CC-BY-4.0); SPRING-INX R1/R2 (SPRING Lab, IIT Madras, CC-BY-4.0); Kathbath (AI4Bharat IndicSUPERB, CC0). Models: hishab (Conformer Large, FastConformer); SayedShaun (Whisper Medium mirror of tugstugi's competition model; wav2vec2 mirror of qdv206's); ehzawad (Phase 1, Phase 4, Qwen3 adapter). Author of this benchmark: Emrul Zawad.